

Question Paper

Exam Date & Time: 01-Jan-2026 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER M.TECH DEGREE MAKEUP EXAMINATIONS - JANUARY 2026

Computational Methods and Stochastic Process [MAT_5128]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) (4)
- A) Apply Gram Schmidt orthogonalization process to construct an orthonormal basis of $\mathbb{R}^3$ from the following linearly independent set $\{(2, -1, 0), (2, -1, 2), (2, -1, -2)\}$.

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) (3)
- Find the QR decomposition of $A = \begin{bmatrix} 3 & 8 \\ 1 & 6 \end{bmatrix}$

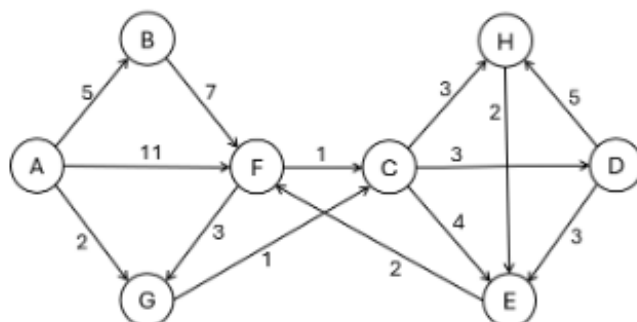
Course Outcome: "CO1" / Cognitive Level: "Application"

- C) (3)
- A certain disease affects 1% of a population.
A diagnostic test gives a positive result for:
- 95% of people who have the disease (true positive rate), and
 - 2% of people who do not have the disease (false positive rate).

If a person tests positive, what is the probability that the person has the disease?

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 2) (4)
- A) Apply Dijkstra's algorithm to find the shortest path from vertex A to all other vertices in the following graph



- B) (3)

Compute u for three-time steps with $h = \frac{1}{4}$,

$$\begin{aligned}\frac{\partial^2 u}{\partial x^2} - 4 \frac{\partial u}{\partial t} &= 0, \\ 0 < x < 1, t > 0, \\ u(x, 0) &= 2x^2 + 1, u(0, t) = u(1, t) = 0.\end{aligned}$$

- C) (3)

Maximize by graphical method, $Z = 50x + 40y$

subject to

$$\begin{aligned}x + 2y &\leq 10, \\ 2x + y &\leq 10, \\ x \geq 0, y &\geq 0.\end{aligned}$$

- 3) (4)

Maximize by simplex method, $Z = 3x_1 + 2x_2$

subject to

A)

$$\begin{aligned}2x_1 + x_2 &\geq 8, \\ x_1 + 2x_2 &\leq 10, \\ x_1, x_2 &\geq 0.\end{aligned}$$

- B) (3)

Use the dominance method to reduce the following game between player A and player B and determine its solution.

	Player B			
Player A	8	10	6	9
	7	8	5	8
	4	3	7	6

- C) (3)

Investor A chooses horizon 1-year, 3-year, or 5-year.

Market can be Downturn, Stable, or Boom.

Expected returns (annualized % for A) are as follows:

- 1-year gives 2% (downturn), 5% (stable), 8% (boom).
- 3-year gives 4% (downturn), 3% (stable), 7% (boom).
- 5-year gives 6% (downturn), 5% (stable), 6% (boom).

Construct the payoff matrix for A and determine whether a saddle point exists. If yes, state the equilibrium and value.

- 4) A researcher evaluates two machine learning models (4)

A)

using 6-fold cross-validation:

Model A	91	89	90	92	88	93
Model B	88	85	87	86	88	89

Test whether there is a significant difference in the mean accuracies of the two models at $\alpha = 0.05$. Assume the two samples are independent and have equal variances.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Suppose that 2 cards are drawn at random from a (3)

pack of cards. Let X be the number of aces and Y be the number of kings obtained. Find joint probability distribution of X and Y .

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) A machine produces items, and each item is defective (3)

with probability 0.2 and non-defective with probability 0.8. Then define Bernoulli's random variable X for item is defective and hence find probability mass function and expectation and variance.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) An urn contains 3 white balls and 2 black balls. The following (4)

A)

experiment is repeated indefinitely: A ball is drawn from the urn. If the ball is white, it is put back in the urn. Otherwise, it is left out. Let X_n be the number of black balls remaining in the urn after n draws from the urn. Find the Transition Probability Matrix of the Markov chain $\{X_n\}$.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Let $\{X_n, n = 0, 1, 2, \dots\}$ be a Markov chain with state (3)

space $\{0, 1, 2\}$ and the initial probability distribution

$P(X_0 = 0) = 0.4, P(X_0 = 1) = 0.4$ and

$P(X_0 = 2) = 0.2$. One step Transition matrix is given by

$$\begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.3 & 0.5 & 0.2 \\ 0.2 & 0.4 & 0.4 \end{bmatrix}$$

Evaluate

(i) $P(X_2 = 1, X_1 = 0 | X_0 = 0)$

(ii) $P(X_3 = 2, X_2 = 1, X_1 = 0, X_0 = 2)$.

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) A machine can be in one of two states - Working (W) or Broken (B). (3)
- If the machine is working this week, the probability it will still be working next week is 0.8.
- If it is broken, the probability it will be repaired and working next week is 0.7. Find the transition probability matrix and determine the long-run performance probabilities.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
058-MIT BLR-MAKEUP EXAM PG-2025-2026	MAT_5128:Computational Methods and Stochastic Process

#	Course Outcome	Description	Marks
1	1	CO 1	7
2	2	CO 2	7
3	3	CO 3	13
4	4	CO 4	13
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM PG-2025-2026	MAT_5128:Computational Methods and Stochastic Process

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	8	26(52%)
4	Analysis	7	24(48%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)